

UPS sizing: Power cuts aren’t just rural

India’s urban tier-II cities still see micro-outages that wreak havoc on switch firmware. Aim for a UPS that keeps core gear alive for 15 minutes – enough to ride out a blip or shut down gracefully. Eaton’s selector suggests adding 25% buffer wattage; APC says keep the load near 80% of UPS capacity for battery longevity.



Quick Calculator:

$$\text{Total Watts (routers + switches + NAS)} \times 1.25 = \text{Minimum UPS Wattage}$$

A 600-W load needs a 750-W UPS; pick a 1 kVA model if you expect to add more PoE later.

At a glance: Indian SMEs can keep every switch, access point and server humming through outages, load spikes and malware probes – provided they have the right-size power reserves and choose a kit with room to grow.

Finish the UPS equation: Runtime, not just watts

Most founders stop at “How many watts can my UPS carry?” The better question is “For how long?” Eaton’s sizing guide recommends adding runtime so staff can close documents and safely

shut down servers. Its worksheet multiplies total load (W) by desired minutes to yield VAh, then matches that to battery packs.

- **Step 1:** List router, switch, NAS and core desktop wattage.
- **Step 2:** Decide on a 15-minute window (enough for India’s typical mini-outage).
- **Step 3:** Enter those numbers into the Eaton calculator or use $\text{Load_W} \times 0.25 = \text{Wh}$; divide by battery voltage to get amp-hours.

A 600 W rack needs roughly 150 Wh; on a 48 V battery string that is 3.1 Ah – well within a 1 kVA tower’s pack. Add 20 % headroom for ageing cells (APC’s own derating rule).

Redundancy matters: N+1 and maintenance bypass

A single UPS is a single point of failure. The classic fix is an N+1 architecture: deploy two identical UPS modules, either in parallel or through a bypass switch, so one can fail without bringing the rack down. APC’s white paper calls this the minimum level of “fault tolerance” before you reach expensive 2N designs. Eaton recommends a maintenance bypass – a simple breaker box that lets you hot-swap batteries or an entire unit.

Do this drill

1. Mount two 1 kVA UPS units side-by-side.
2. Feed each from a separate wall circuit where possible.
3. Run both outputs into an automatic transfer switch (₹6–10 k on Indian e-commerce sites).
4. Pull the plug on UPS A – your gear should stay on UPS B with zero downtime.

For larger offices, consider a 2N+1 ring where each UPS line is itself N+1; financial data centres use it, but SMEs can scale down with three 2 kVA units: two active, one spare.

Don’t forget surge & earthing

More electronics die from bad earth loops than from headline-grabbing lightning. Cisco’s power-requirement reference reminds

Tier	Router	PoE Switch	Wi-Fi AP	UPS	Ideal For
Bootstrapped (≤ ₹20 k)	Dual-WAN ARM router, 900 Mbps NAT, IPSec 200 Mbps (e.g., GLiNet Flint 2)	8-port L2 PoE+ 55 W budget	1× Wi-Fi 6 dual-band 2×2 AP (draw ≈12 W)	1 kVA line-interactive	Solo founder or 5-seat studio
Balanced (₹20–60 k)	Cisco RV 345-P: dual WAN, hardware AES, 100 VPN tunnels	8-port PoE+ 125 W, SFP uplinks, basic L3 static routes	2× Wi-Fi 6E tri-band APs, 4×4 radios, 22.5 W draw (EnGenius ECW336)	2 kVA online UPS + bypass	20-seat SaaS office, hybrid work
Growth-Ready (₹60–150 k)	x86 firewall appliance, 2.5 GbE, WireGuard 1 Gbps, redundant PSU	24-port PoE++ 370 W, Layer 3 dynamic routing, stacking	4× Wi-Fi 6E APs w. DFS scan & WPA3-Enterprise (Grandstream GWN7665)	Dual 2 kVA UPS in N+1, SNMP card	50-seat co-working, IoT devices, guest portal